

TRUEVALUE ANALYTICS · INTERNAL WORKING DOCUMENT

From Illustrative to Investor-Ready: What We Still Need to Know

A gap analysis and research agenda identifying every claim, score, projection, and assumption in the Investor Report that requires real data, validation, or policy research before the report can be presented without qualification to institutional investors.

Document Type

Gap Analysis / Research Agenda

Subject Report

investor_report.html (March 2026)

Audiences

Non-technical & Technical

Status

Working Draft — Internal Only

This document is a frank internal assessment of what is currently synthetic, estimated, or unvalidated in the Investor Report. It is not for external distribution. Its purpose is to create a clear research agenda for moving the report from a structural demonstration to an evidenced institutional document.

HOW TO READ THIS DOCUMENT

Two Audiences, One Document

This report is written for two readers simultaneously. **The non-technical reader** — a business associate, a potential partner, or a stakeholder who needs to understand what questions remain unanswered — will find plain-language questions in Section 2, organised by theme. No technical background is required to understand or act on them.

The technical reader — the analyst or developer who needs to know exactly which data fields are synthetic, which formulas are incomplete, and which calculations contain internal inconsistencies — will find a precise field-level audit in Section 3. This section references specific variables, file paths, and formula definitions by name.

Both audiences share the same goal: understanding what needs to be done before the Investor Report can be presented to a sophisticated institutional investor who will ask hard questions.

Status codes used throughout: ✓ **REAL / CITED** ♦ **STRUCTURALLY DERIVED**

~ **ESTIMATED / MODELLED**

✗ **SYNTHETIC**

⚠ **INTERNAL INCONSISTENCY**

4 REAL / CITED Claims grounded in publicly verifiable, sourced data (LBMA structure, COMEX, Foran 2005 TBL benchmarks, Shea programme documentation)	6 STRUCTURALLY DERIVED Claims about industry structure that are factually accurate but lack a specific citation that would survive formal due diligence	5 ESTIMATED / MODELLED Numbers derived from a logical model or analogy — defensible in principle but not measured from real supply chain data	12 SYNTHETIC Numbers explicitly generated from a structural model using assumed parameters. The report's Appendix discloses this, but it limits the report's evidential weight.
--	--	--	--

⚠ **One internal inconsistency identified:** The balance scores displayed in the Investor Report do not match the formula stated in its own Appendix ($\text{Balance Score} = \frac{\min(C, G)}{\max(C, G)} \times 100$). For example, Phase 6 Logistics shows C=380, G=175, which by the stated formula gives 46.1% — but the report displays 37%. This discrepancy appears across all nine phases and needs to be resolved before the report circulates to any

technically sophisticated reader. See Section 3, Item T-1 for full detail.

CURRENT STATE AUDIT

What Each Section of the Investor Report Is Actually Based On

The table below maps every substantive section of the Investor Report to its current evidential status. This is the master reference from which both the non-technical questions and the technical requirements below are derived.

REPORT SECTION	KEY CLAIMS / DATA USED	CURRENT STATUS	WHAT CHANGES IT
Executive Summary <i>KPI cards: 37%, 3/9, +71%, 91%</i>	All four headline KPI figures are derived from the synthetic phase dataset	✗ SYNTHETIC	Real C/G component data from at least one pilot participant in each critical phase
Section 1 — Transparency Table <i>Phase opacity descriptions</i>	Phase names, transparency ratings, and opacity reasons are derived from LBMA and COMEX structure — factually accurate descriptions of industry design	◆ STRUCTURAL	Add specific citations (LBMA Responsible Gold Guidance, OECD Due Diligence) to each row
Section 2 — C/G/N Formula <i>Balance Score formula; Net Performance; Sustainability Index</i>	Balance Score formula disclosed. However: (a) "Balance Factor" in Net Performance is undefined; (b) Sustainability Index formula is described verbally but not mathematically; (c) C and G sub-components are not disclosed anywhere in the report	~ PARTIAL	Define Balance Factor precisely; publish complete Sustainability Index formula; disclose at least the category-level C/G sub-components

REPORT SECTION	KEY CLAIMS / DATA USED	CURRENT STATUS	WHAT CHANGES IT
Section 3 — Gold Phase Scores <i>All 9 phase C, G, balanceScore values</i>	All 27 values (9 phases × C, G, score) are synthetic integers seeded from structural reasoning, not measured from real data	✗ SYNTHETIC	Real data from even one LBMA mining company (Phase 1) and one COMEX report (Phase 7) would allow partial replacement; Phases 3, 4, 6 require partner negotiation
Section 3 — "90% of gold refined by 5 facilities"	Widely cited industry fact but requires a current citation; the LBMA Good Delivery list has evolved	◆ STRUCTURAL	Verify current LBMA accredited refiner market share data; cite WGC Gold Supply report year
Section 4 — Shea Phase Scores <i>SHEA.constraints=215, SHEA.growth=205, balanceScore=91</i>	These values are attributed to "Serious Shea / Acorn programme operational data" in the Appendix, but are not confirmed as actual measurements from those programmes. Also: applying the stated formula to C=215 / G=205 gives 95.3%, not 91%.	⚠ INCONSISTENCY	Confirm C and G values directly with Serious Shea or Acorn; reconcile the stated formula with the displayed score
Section 4 — "Women's Gold" & Shea narrative	The "Women's Gold" characterisation and cooperative structure descriptions are consistent with Mirova and Serious Shea documentation	✓ REAL	Add specific citation to Mirova feasibility report or Acorn programme documentation
Section 5 — TBL Optimization Scores <i>11 indicator current/optimised balance score estimates</i>	All 22 balance score values (11 current + 11 optimised) are analytical estimates based on the Foran 2005 TBL data and	~ ESTIMATED	These are the most defensible estimates in the report because they derive from peer-reviewed

REPORT SECTION	KEY CLAIMS / DATA USED	CURRENT STATUS	WHAT CHANGES IT
	TrueValue framework logic — they are not measured		Foran data; they need framing as "analytical estimates" rather than measurements
Section 5 — Water/ biodiversity credit estimates "\$20–50M/yr" and "\$5–30M/yr"	These dollar ranges are estimated from knowledge of water trading and biodiversity credit markets but are not cited to specific transactions or market rate schedules	~ ESTIMATED	Cite specific water market rates (Australian Water Register, Murray-Darling Basin Authority) and biodiversity credit scheme documentation
Section 6 — 10-Year Projection (+71%) <i>Extractive and TrueValue growth rate series</i>	Growth rates (7% → 4% → 1.5% extractive; 6.5% → 9% → 11.5% TrueValue) are "derived from structural efficiency model" — which means they are assumed. No empirical grounding exists.	✗ SYNTHETIC	Anchor growth rate differentials to at least one empirical analogy: a supply chain reform that produced measurable financial improvement (Fairtrade premium data; CDP disclosure adoption studies; LBMA RGG adoption outcomes)
Section 7 — Regulatory alignment value <i>"Future disclosure requirements will force balance..."</i>	Factually well-grounded (EU CSRD, SEC climate rule, OECD DDG are all moving in this direction) but no specific regulation is named	◆ STRUCTURAL	Name EU CSRD Article 29a, SEC Climate Disclosure Rule 33-11275, and OECD DDG 5th Edition specifically
Appendix — Data Transparency Notice	The disclosure that data is synthetic is clear and honest. However "national accounts" is cited as a source basis	◆ STRUCTURAL	Replace generic "national accounts" with specific Foran et al. 2005 citation (already in the

REPORT SECTION	KEY CLAIMS / DATA USED	CURRENT STATUS	WHAT CHANGES IT
	without specifying which national accounts (Foran 2005 is the clear reference).		Phase Architecture Sources list — should be cross-referenced here)

SECTION 2 — FOR THE NON-TECHNICAL READER

The Plain-Language Questions We Need Answered

These are the questions a well-informed non-technical associate — or a non-specialist investor — would ask if they read the Investor Report carefully. They are organised by theme. Each question identifies a gap that, once resolved, directly strengthens a specific claim in the report.

THEME A: THE CORE SCORES — WHERE DO THE NUMBERS COME FROM?

A1 **The balance scores (37%, 54%, 61%) are the centrepiece of the investor argument. Are they real measurements, or are they calculated from a model?**

DATA *Currently synthetic. The Appendix discloses this honestly. To make them real, we need C and G component data from at least one participant in each phase. What would we need to ask a mining company, refinery, or logistics operator for?*

A2 **The report defines Constraints and Growth Capacity but never lists exactly what goes into them. What are the specific factors that make up, say, the Logistics phase Constraints score of 380? Could we explain this to an investor?**

VALIDATION *This is the most important question for investor credibility. If we cannot show the sub-components, a sophisticated investor cannot validate the scores independently — which is a red flag.*

A3 **We compare Shea's Phase 6 (91% balance) to Gold's Phase 6 (37%). Is the 91% score measured from actual Serious Shea or Acorn data, or is it also modelled?**

DATA **SHEA** *This comparison is the emotional heart of the Shea section. If both scores are synthetic, the comparison loses its force. Do we have a contact at Serious Shea or Acorn who could validate or provide the underlying numbers?*

A4 **We have set target balance scores for each phase (e.g., 85% for Mining, 50% for Logistics). How were these targets decided? Is there a reference for what a "good"**

target looks like in each phase?

VALIDATION *These targets shape how the whole scoring system is interpreted. If they were set by judgment alone, they need at minimum a documented rationale. At best, an analogy from another framework (Fairtrade certification standards, LBMA responsible sourcing thresholds) would validate them.*

THEME B: THE GOLD CHAIN — WHAT DO WE ACTUALLY KNOW?

B1 **We state that approximately 90% of the world's newly mined gold is processed by five LBMA-accredited refineries. Is this number current? The LBMA Good Delivery list has more than five refineries on it — do we mean five specific dominant ones?**

DATA *The WGC Gold Demand Trends report and LBMA Annual Report should have current refinery throughput data. We need a year-stamped citation.*

B2 **We describe refinery fee schedules as "private" and "secret." Do we know even a ballpark range? Even a broad range (\$2–8/oz?) would make the "hidden cost" claim more concrete without requiring disclosure of confidential data.**

DATA *Publicly disclosed mining company financial reports sometimes contain treatment charge ranges. Court cases involving refinery disputes occasionally reveal fee structures. Industry analysts (Wood Mackenzie, Metals Focus) sometimes publish estimated ranges.*

B3 **We describe the Logistics & Vaulting phase as "structurally architected for opacity." Has anyone on the team spoken directly with a vault operator, security logistics professional, or insurance broker who can confirm this characterisation from the inside?**

VALIDATION *A single attributed or anonymised firsthand account from a logistics professional would transform this from a structural inference into a validated finding. Former Brink's or G4S employees, Lloyd's of London precious metals underwriters, or bullion banking risk officers are accessible sources.*

B4 **What volume of gold actually flows through each of the nine phases annually? We describe the phases qualitatively but give no physical tonnage data. Could we add approximate volumes (using WGC supply statistics) to make the scale of each phase tangible?**

DATA *The WGC publishes annual mine production (~3,600 t), recycling (~1,200 t), and COMEX/LBMA volumes. These would make the transparency table and phase descriptions significantly more concrete.*

THEME C: THE SHEA STORY — DO WE HAVE THE DATA?

C1 **We cite "Serious Shea / Acorn programme operational data" as a source for the Shea balance scores. Have we actually received data from them, or are these scores estimated based on their published documentation?**

SHEA DATA This is the most important Shea question. If the data came directly from Serious Shea or Acorn, the comparison to gold gains enormous credibility. If it was estimated from their public documents, that should be stated and a direct data request made.

C2 We describe the Shea phase 6 structure as: "Cooperative network, traceable custody, multiple buyers." Approximately how many buyers does a typical Serious Shea cooperative have access to? And how many farmers are in the cooperative network today?

SHEA Specific numbers (e.g., "cooperatives in Burkina Faso and Senegal with ~85,000 enrolled women farmers across 12 buyer relationships") would make the Shea comparison dramatically more powerful and credible.

C3 We list "Burkina Faso / Senegal value chain studies" in the Appendix. Do we have those specific studies filed, and can we name them precisely (title, author, year)?

SHEA VALIDATION Generic citations like "value chain studies" weaken the Appendix credibility. The Mirova Senegal Feasibility Report and the Acorn MRV documentation are likely the specific sources — they should be named.

C4 The report says shea farmers receive below-average income compared to the supply chain value they generate. Do we have a specific figure — what does a shea picker earn per day or per season, and what is the value of the shea butter at the point of sale to end buyers?

SHEA DATA The income inequality across the shea chain is a powerful investor story. A specific data point (e.g., "collector earns \$0.40/kg; retail shea butter sells for \$12/kg") would make it concrete. Mirova and GIZ both publish this kind of data.

THEME D: THE 10-YEAR PROJECTION — IS +71% DEFENSIBLE?

D1 The projection showing a +71% cumulative TrueValue advantage at Year 10 is the most compelling number in the report. The Appendix honestly discloses that the growth rates come from a structural model, not from historical data. Is there any real-world supply chain case we can point to where a transparency-driven reform produced a measurable financial improvement?

VALIDATION Fairtrade adoption data, CDP score improvements correlated with cost of capital, LBMA RGG adoption and subsequent gold price premium, FSC certification and timber price premiums — any of these could serve as an empirical anchor for the projection's direction, even if not the exact magnitude.

D2 The model assumes the extractive path grows at 7% initially but declines to 1.5% by Years 7–10. What is the reasoning for this specific deceleration? Would an investor's economist accept this without a documented rationale?

VALIDATION The deceleration story is actually very plausible (accumulating constraint costs) but needs a narrative justification with named mechanisms, not just assumed rates. What specific events trigger the slowdown — regulatory action, supply disruption, reputational event? These should be written out explicitly.

- D3** The projection uses a single base index of 100 for both paths. Investors will ask: 100 of what? Is this revenue? EBITDA? Asset value? Equity returns? The report doesn't specify, which makes the +71% figure harder to interpret.

VALIDATION Defining the base metric (even as "illustrative value index") and explaining whether it represents total return, equity return, or supply chain operational value would make the projection more interpretable without requiring real financial data.

THEME E: THE FRAMEWORK ITSELF — CAN IT BE CHALLENGED?

- E1** Has anyone outside our team reviewed the TrueValue methodology? An expert validation — even a letter of review from a university supply chain professor or a sustainability research institute — would significantly raise the report's credibility for institutional investors who perform due diligence.

VALIDATION Suggested reviewers: CSIRO Sustainable Ecosystems (the Foran et al. institution); the University of Sydney Business School; the World Resources Institute; Chatham House Resource Trade Earth programme.

- E2** Are there other supply chain balance or transparency scoring frameworks in the market that we should acknowledge, differentiate from, or potentially partner with? Investors will ask whether TrueValue is unique.

POLICY Known potential comparators: EcoVadis supply chain scoring; CDP supply chain programme; Thomson Reuters Supply Chain Risk scores; Sourcemap traceability platform; the forthcoming EU CSRD supply chain due diligence metrics. A brief competitive landscape paragraph in the Appendix would address this.

- E3** Is the TrueValue methodology IP-protected? Before this report circulates more widely, should we register a trademark, file a methodology patent, or at minimum establish a clear terms-of-use for the framework's use by third parties?

POLICY The current report is copyright TrueValue Analytics 2026, but the methodology itself (the C/G balance formula, the phase architecture) is not yet IP-protected as far as this document indicates. This should be a legal consultation item before any broad distribution.

THEME F: REGULATORY & MARKET CLAIMS — CAN WE BE SPECIFIC?

- F1** We say "future disclosure requirements will force balance towards TrueValue targets." Which regulations specifically? Naming them — EU CSRD, SEC climate rule, OECD Due Diligence Guidance — turns a vague prediction into a documented regulatory trajectory.

POLICY The EU Corporate Sustainability Reporting Directive (CSRD), effective 2024–2026 rollout, requires supply chain due diligence. The SEC's climate disclosure rule (33-11275) requires scope 3 emissions disclosure. OECD's DDG 5th Edition specifically covers mineral supply chains. All three directly mandate the kind of phase-level data TrueValue provides.

- F2** We claim "ESG-driven capital reallocation increasingly targets chains with

undisclosed custody arrangements." What is the actual scale of ESG assets currently avoiding or underweighting opaque gold supply chains? A number here — even an approximate one — makes the claim investable rather than rhetorical.

DATA The GFANZ (Glasgow Financial Alliance for Net Zero) tracks commodity ESG commitments. BloombergNEF and Morningstar publish ESG fund flow data. A search for "gold supply chain ESG exclusion" in institutional investor ESG policy disclosures would find specific fund exclusion criteria.

F3 The Optimization Analysis section estimates water credits worth \$20–50M/year and biodiversity credits worth \$5–30M/year for Australian gold mining. Are there actual transactions in these markets for mining-adjacent land that we can cite, or are these purely theoretical market-rate estimates?

DATA The Australian Carbon Credit Unit (ACCU) scheme and the emerging Australian Biodiversity Offset Programme (ABOP) have published transaction data. Any confirmed transaction where a mining company received water or biodiversity credit income would validate the range.

SECTION 3 — FOR THE TECHNICAL READER

Field-Level Data Requirements and Formula Gaps

This section catalogues every specific variable, formula, and data object in `investor_report.html` that is currently synthetic, undefined, or internally inconsistent. Each item includes the variable name as it appears in the source code, its current value, the gap, and what resolution looks like.

T-1 · CRITICAL: Balance Score Formula vs. Displayed Values ⚠️ INTERNAL INCONSISTENCY

The Appendix states: $\text{Balance Score} = \min(C, G) / \max(C, G) \times 100$. Applying this formula to the phase data objects reveals that the displayed balance scores are *systematically lower* than the formula would produce. This is not a rounding issue — the gap is 6–13 percentage points across all phases. Either the formula in the Appendix is incomplete (there is an additional factor not disclosed), or the balance scores and C/G values were set independently of each other.

Phase	C / G values	Formula gives	Report shows	Gap
Ph.0	C=195, G=148	75.9%	68%	–7.9pp
Ph.1	C=330, G=316	95.8%	90%	–5.8pp
Ph.4	C=340, G=210	61.8%	54%	–7.8pp

Ph. 6	C=380, G=175	46.1%	37%	-9.1pp — largest gap
SHEA	C=215, G=205	95.3%	91%	-4.3pp

Resolution: Either (a) define the complete Balance Score formula including the hidden factor, update the Appendix formula accordingly, and verify all displayed scores are recalculated consistently; or (b) recalculate all balance scores using the stated formula and update the phase data object and all downstream displays.

T-2 · C and G Sub-Component Definitions

X UNDEFINED

The `PHASES[]` array stores `constraints` and `growth` as single integers (e.g., `constraints:380` for Phase 6). No sub-component definition exists anywhere in the codebase or documentation. Before the report can withstand investor due diligence, the following must be defined for each phase:

C_sub[]	What specific factors constitute the Constraints score? (regulatory burden, contract rigidity, counterparty concentration, jurisdictional exposure, capital lockup, opacity premium) — minimum 4–6 named factors per phase with individual weights	X MISSING
G_sub[]	What specific factors constitute the Growth Capacity score? (market access, alternative counterparties, traceability capability, innovation pathway, pricing discovery) — minimum 4–6 named factors per phase with individual weights	X MISSING
balanceTarget	Values [75, 85, 85, 80, 80, 85, 50, 85, 80] have no documented basis. Need explicit rationale for each, or derivation from an analogous framework threshold	X UNDOCUMENTED

T-3 · Net Performance Formula — Undefined Balance Factor

~ INCOMPLETE

The Appendix states: $\text{Net Performance} = \sqrt{(\text{Constraints} \times \text{Growth Capacity}) \times \text{Balance Factor}}$. The term "Balance Factor" is not defined anywhere in the report, codebase, or data files.

Balance Factor	Is this equal to (Balance Score / 100)? Or is it a separate normalisation term? Its definition determines whether N is simply the geometric mean scaled by the balance score, or something more complex. This must be formalised before the methodology can be presented as complete.	X UNDEFINED
Sustainability Index	Described as "inverse of the squared imbalance plus a phase-specific energy base cost." This is not a complete mathematical expression. What is the exact formula? What is "phase-specific energy base cost"? How is it measured? The value 0.91 for Shea and 0.05 for Gold Ph.6 need to be reproducible from a stated formula.	X INCOMPLETE

T-4 · SHEA Object — Source and Consistency		⚠ NEEDS VERIFICATION
SHEA.constraints	Value: 215. Source attributed to "Serious Shea / Acorn operational data" but no specific document or data request confirmed. Verify: is this from Acorn MRV data, Mirova feasibility model, or estimated by the TrueValue team?	X UNCONFIRMED
SHEA.growth	Value: 205. Same verification need as above.	X UNCONFIRMED
SHEA.balanceScore	Value: 91. Does not match $\min(215,205)/\max(215,205) \times 100 = 95.3\%$. Same formula inconsistency as T-1. Needs reconciliation.	⚠ INCONSISTENCY
SHEA_PH6.transparency	Radar chart normalised value: 68 (out of 100). Basis for this normalisation is not documented. How was "Medium-High" transparency converted to the value 68?	X UNDOCUMENTED
SHEA_PH6.growthIdx	Radar chart value: 95. Not derived from the SHEA.growth score of 205. What normalisation or scaling was applied?	X UNDOCUMENTED

SHEA_PH6.stability

Radar chart value: 87. This metric does not appear elsewhere in the framework definition. What is "Phase Stability" and how is it calculated?

X UNDEFINED METRIC**T-5 · Projection Growth Rates — Assumed, Not Modelled****X ASSUMED**

The projection chart uses two hardcoded rate arrays. Neither has a documented derivation.

extractiveRates[]

Values: [0, 0.07, 0.07, 0.07, 0.04, 0.04, 0.04, 0.015, 0.015, 0.015, 0.015]. The deceleration from 7% to 4% to 1.5% reflects accumulating constraint costs — the direction is logical, but the specific values need: (a) a documented rationale, or (b) anchoring to an empirical case of supply chain degradation (e.g., cost escalation in an opaque mining chain)

X ASSUMED**truevalueRates[]**

Values: [0, 0.065, 0.065, 0.065, 0.09, 0.09, 0.09, 0.115, 0.115, 0.115, 0.115]. The acceleration from 6.5% to 9% to 11.5% represents the TrueValue compounding advantage — also logical directionally, but needs empirical anchoring (e.g., Fairtrade premium adoption data, CDP study showing disclosure → lower cost of capital)

X ASSUMED**Base index definition**

The "base index = 100" is not defined as representing a specific financial metric (revenue, EBITDA, equity return, asset value). This must be specified before the projection can be formally interpreted.

X UNDEFINED**T-6 · Section 5 Optimization Balance Scores****~ ESTIMATED**

The 11 "current" and 11 "optimised" balance scores in `OPTIM_DATA[]` are analytical estimates grounded in the Foran 2005 TBL data. They are the most defensible estimates in the report (because they derive from peer-reviewed data) but need explicit framing as estimates, and the optimised scores need engineering or economic validation.

current scores (11)	Values [45,50,55,68,72,70,40,50,50,65,42]. These are analytical estimates applying TrueValue C/G logic to TBL intensity ratios. Each score needs a documented derivation showing how the TBL intensities map to C/G component weights.	~ ESTIMATED
optimised scores (11)	Values [78,82,80,88,91,90,65,78,75,88,70]. The optimised GHG score (88%) implies 44% electricity source reduced to ~15%. Needs engineering validation: what renewable energy penetration rate, at what cost, achieves this? This is the most likely score to be challenged.	~ NEEDS VALIDATION
water credit estimate	"\$20–50M/yr uncaptured." Needs specific citation to Australian water trading market rates applied to a stated hectare-equivalent or ML-equivalent water efficiency baseline.	~ NEEDS CITATION

T-7 · Claims Requiring Specific Citations**◆ STRUCTURAL — NEEDS CITATIONS**

"~90% refined by 5 facilities"	Add: WGC Gold Supply report year; LBMA Good Delivery List current accredited refiner market share data	◆ ADD CITATION
EU/SEC/OECD regulation	Section 7: name EU CSRD Article 29a (supply chain due diligence), SEC Rule 33-11275 (Scope 3 disclosure), OECD DDG 5th Edition (mineral supply chains)	◆ ADD CITATION
Foran 2005 cross-reference	Appendix Data Transparency Notice cites "national accounts" generically — should cross-reference to the Foran et al. (2005) entry already in the Phase Architecture Sources list	◆ ADD CROSS-REF
Shea Mirova citation	Appendix lists "Mirova feasibility research" — should be: Mirova Natural Capital (2021), "Shea Agroforestry: Feasibility Assessment, Senegal" (or equivalent full title)	◆ SPECIFY

SECTION 4 — PRIORITY MATRIX

What to Tackle First: Three Tiers of Urgency

Not all gaps are equal. The priority matrix below separates the items that would immediately undermine the report's credibility with a sophisticated investor (Tier 1) from those that significantly strengthen it (Tier 2) and those that represent longer-term quality improvements (Tier 3). Tier 1 items should be resolved before the report leaves this team.

TIER 1 — FIX BEFORE EXTERNAL DISTRIBUTION	TIER 2 — SIGNIFICANTLY STRENGTHENS CREDIBILITY	TIER 3 — LONG-TERM QUALITY
T-1: Balance Score Formula The formula in the Appendix does not produce the displayed scores. A technically literate investor will spot this immediately. Resolve the discrepancy or explain the full formula.	T-3: Complete the Formulas Define Balance Factor precisely; formalise the Sustainability Index as a mathematical expression.	B2: Refinery Fee Ranges Obtain even approximate treatment charge ranges from industry analysts or court records to make the "hidden cost" claim concrete.
T-2: C/G Sub-Components Without a sub-component definition, the scores cannot be independently verified. Prepare at least a one-page methodology annex defining the factors.	B1: Verify 90% Refinery Claim Get a current WGC citation for LBMA refinery market concentration.	B4: Physical Volume Data Add annual tonnage flowing through each phase using WGC supply statistics.
A3/T-4: Shea Data Source Confirm whether the Shea C/G scores (215/205) came from Serious Shea or were estimated. If estimated, acknowledge this in the report.	D1: Empirical Projection Anchor Find one real-world case (Fairtrade, CDP, LBMA RGG) where structural reform produced measurable financial improvement.	C2/C4: Shea Farmer Data Obtain specific cooperative size, buyer count, and farmer income data from Serious Shea or GIZ.
D3/T-5: Define the Base Index Specify what the projection's "Value Index = 100" represents (revenue, equity, asset value) so the +71% has an interpretable meaning.	F1: Name Specific Regulations Replace the vague "future disclosure requirements" with EU CSRD, SEC 33-11275, and OECD DDG 5th Edition.	F2: ESG Capital Flow Data Quantify the scale of ESG assets currently excluding or underweighting opaque gold supply chains.
E3: IP Status Before broader distribution, confirm the methodology has appropriate legal protection or an explicit open-source licence.	C3: Specify Shea Citations Replace generic "Burkina Faso / Senegal value chain studies" with full Mirova and Acorn document titles.	F3: Credit Market Citations Cite specific Australian water and biodiversity credit market transactions to validate the \$25–80M/yr estimate.
	E1: Expert Review Seek a letter of methodological review from CSIRO, University of Sydney, or World Resources Institute.	E2: Competitive Landscape Add a brief paragraph to the Appendix acknowledging comparable frameworks (EcoVadis, CDP supply chain, Sourcemap) and distinguishing TrueValue from each.

SECTION 5 — SUGGESTED NEXT STEPS

A Practical Action Plan

The gaps identified in this document are not an indictment of the Investor Report — they are the normal condition of a first-generation analytical framework that has been built rigorously from the structural logic of its subject matter. The report is intellectually honest (the Appendix clearly discloses the synthetic data basis) and structurally sound. What it now needs is a targeted evidence-gathering effort to replace synthetic values with real ones, confirm its formulas are internally consistent, and anchor its projections in at least one empirical analogy.

Estimated effort to reach Tier 1 resolution: Most Tier 1 items can be resolved by one analyst in 4–6 weeks without requiring external data access. The formula inconsistency (T-1), sub-component definitions (T-2), and projection base index (D3/T-5) are all internal documentation tasks. The Shea data confirmation (A3/T-4) requires one outreach email to Serious Shea or Acorn. IP status (E3) requires one legal consultation.

IMMEDIATE ACTIONS (WEEKS 1–4)

- Resolve the Balance Score formula discrepancy (T-1) — either fix the displayed scores or document the complete formula
- Write a one-page C/G sub-component definition document for at least Phases 3, 4, and 6 (the three critical phases)
- Contact Serious Shea / Acorn to confirm the status of the Shea C/G data
- Define the Sustainability Index mathematically and add it to the Appendix
- Define the projection base index metric (revenue, equity return, or operational value)

SHORT-TERM RESEARCH (WEEKS 4–12)

- Verify LBMA refinery market concentration data with a WGC citation
- Name specific EU CSRD, SEC, and OECD regulations in Section 7
- Identify one empirical analogy for the projection (Fairtrade premium data or CDP cost-of-capital study)
- Obtain full citations for Mirova and Acorn source documents
- Draft a brief competitive landscape paragraph for the Appendix
- Consult on IP protection for the TrueValue methodology

MEDIUM-TERM VALIDATION (MONTHS 3–6)

- Approach CSIRO or University of Sydney Supply Chain Research for a methodology review letter
- Engage one major mining company as a pilot data participant (Phase 1 data)
- Develop contact at Brink's, G4S, or a Lloyd's bullion underwriter for Phase 6 qualitative validation
- Quantify the ESG capital flow claim with a Bloomberg or Morningstar data point

DATA COLLECTION TARGETS (MONTHS 3–12)

- WGC Gold Supply annual report — Phase 1 production volumes and LBMA refiner market share
- Acorn MRV documentation — Shea C/G input data confirmation
- Metals Focus or Wood Mackenzie — estimated treatment charge ranges for doré refining
- Australian Water Register — water trading market rates for mining-adjacent land
- GIZ or CIFOR — Shea farmer income data (collector price vs end-buyer price)

APPENDIX

Reference: Current Data Status by Variable

VARIABLES CONFIRMED AS REAL / CITED

- Phase names and sequence (0–8): standard industry structure per LBMA and WGC
- GHG source split (44% electricity, 13% direct, etc.): Foran et al. 2005, Sector 13140010+13170010
- All Foran 2005 TBL intensity ratios (water, land, employment, surplus): peer-reviewed, cited
- Australian gold production (260 t/yr), reserves (5,165 t): Foran 2005, USGS-consistent
- Phase transparency descriptions: structurally derived from LBMA, COMEX regulatory architecture
- Shea "Women's Gold" characterisation: consistent with Mirova and Acorn documentation

VARIABLES CONFIRMED AS SYNTHETIC

- All PHASES[0–8].constraints values (195–380)
- All PHASES[0–8].growth values (148–316)
- All PHASES[0–8].balanceScore values (37–90)
- SHEA.constraints (215), SHEA.growth (205), SHEA.balanceScore (91)
- Radar chart: GOLD_PH6 and SHEA_PH6 transparency/stability/growthIdx values
- extractiveRates[] and truevalueRates[] projection arrays
- All OPTIM_DATA[].current and .optimised balance score estimates

FORMULAS WITH OUTSTANDING GAPS

- `Balance Score` — stated formula does not reproduce displayed values
- `Balance Factor` in Net Performance — undefined term
- `Sustainability Index` — verbal description only; no mathematical expression
- `SHEA_PH6.stability` — metric not defined anywhere in the framework
- `balanceTarget[]` — values undocumented; no derivation stated

EXTERNAL ORGANISATIONS TO CONTACT

- **Serious Shea / Acorn:** Shea C/G data confirmation (Priority: Tier 1)
- **WGC:** current LBMA refiner market share; annual supply data
- **CSIRO / Univ. Sydney:** methodology peer review
- **Mirova Natural Capital:** full Shea feasibility report citation
- **Metals Focus / Wood Mackenzie:** doré refining treatment charge ranges
- **Australian Water Register / ABOP:** water and biodiversity credit rates
- **GIZ / CIFOR:** Shea farmer income data

TrueValue Analytics — Internal Working Document. This gap analysis is intended to support the evidence-strengthening process for the Investor Report. It is not for external distribution. All references to "synthetic" data reflect the Investor Report's own Appendix disclosure. © 2026 TrueValue Analytics.